THERMAL FAULT DIAGNOSTIC REPORT

Automated Infrared Thermographic Analysis & Deep Learning Evaluation

Analysis ID:	f0271bdc-ef06-4b65-ba29-1d1b8bf8225f	**Timestamp:**	2026-10-14T13:33:54
Image Filename:	valid_scan.png	**Model Version:**	ResNet18-ThermalTransfer (v1.2.0-trained)
Condition:	**Fault Detected**	**Fault Class:**	**Insulation Fault**
Confidence Score:	57.8%	**Severity Level:**	**None**

Thermographic Inspection Imagery

Figure 1: Original Thermal Image

Figure 2: AI Diagnostic Overlay (ROI & Hotspot)

Engineering Interpretation & Recommendations

Summary: Normal thermal dissipation observed across equipment body. No abnormal hotspot detected.

Next Actions: Schedule immediate physical inspection of identified terminal contacts. Verify torque on bolted electrical connections, inspect dielectric insulation, and measure phase current balance.

LIMITATIONS & DISCLAIMER:

- Severity levels are derived from thermal gradient contrast and geometry. Uncalibrated digital image formats (PNG/JPG) do not provide calibrated radiometric sensor data in degrees Celsius.
- Solar reflection, airflow, and surface emissivity affect infrared measurements.
- **Automated analysis supports inspection and should not replace qualified maintenance assessment.**